TEACHING LANGUAGE MODELS TO HEDGE HONESTLY: THE CHALLENGE OF ALIGNING LINGUISTIC UNCERTAINTY WITH INTERNAL CONFIDENCE

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ABSTRACT

Large language models frequently express unwarranted certainty in their responses, leading to user overreliance on incorrect outputs. While prior work has focused on calibrating explicit numerical confidence scores, we investigate a more natural approach: training LLMs to express appropriate linguistic uncertainty through hedging language (e.g., "I think" or "probably") that reflects their internal confidence. We introduce Uncertainty-Aligned Hedging (UAH), a reinforcement learning framework that rewards alignment between hedging intensity and token-level entropy. Our experiments on GPT-2 across four QA datasets reveal a fundamental tension: while UAH achieves high absolute alignment scores (0.92 average), the correlation between hedging and entropy remains weak and unstable (Spearman ρ ranging from -0.20 to 0.10). Calibration analysis shows the model learns a compressed hedging range rather than true sensitivity to uncertainty. These negative results highlight that achieving natural, calibrated uncertainty communication in LLMs is substantially harder than optimizing simple alignment metrics.

1 INTRODUCTION

Language models increasingly serve as information sources for consequential decisions, yet they frequently express unwarranted certainty even when their internal probability distributions indicate substantial uncertainty (Zhou et al., 2023). This mismatch between expressed and actual confidence leads to user overreliance on incorrect outputs, with recent work demonstrating this phenomenon persists across languages and cultural contexts (Rathi et al., 2025). Prior approaches have focused on eliciting explicit numerical confidence scores (Tian et al., 2023) or calibrating verbalized confidence through reward modification (Leng et al., 2024). However, humans naturally communicate uncertainty through hedging language—expressions like "I think," "probably," and "I'm not sure"—rather than numerical probabilities.

We introduce Uncertainty-Aligned Hedging (UAH), a fine-tuning framework using reinforcement learning to align hedging intensity with internal model uncertainty measured via token-level entropy. Our experiments reveal sobering results: while UAH achieves high alignment scores (averaging 0.92), deeper analysis exposes fundamental limitations including weak hedging-entropy correlations (Spearman ρ between -0.20 and 0.10), a compressed hedging range clustered around mid-values, and systematic under-hedging at high uncertainty. These findings suggest current alignment approaches may optimize surface-level metrics without achieving genuine uncertainty-aware communication.

2 RELATED WORK

Zhou et al. (2023) demonstrated that LLMs are highly sensitive to epistemic markers, with accuracy varying by over 80% depending on uncertainty expressions—critically, this reflects mimicking rather than true uncertainty reasoning. Leng et al. (2024) showed RLHF causes verbalized overconfidence and proposed reward calibration methods, while Tian et al. (2023) demonstrated that verbalized confidence can be well-calibrated when properly elicited. Zhang et al. (2025) proposed

Table 1: Cross-dataset evaluation. UAH achieves high alignment but weak correlations, indicating surface-level optimization rather than genuine uncertainty tracking.

Dataset	Alignment	Pearson r	Spearman ρ	Baseline
Synthetic QA	0.919	-0.111	-0.203	0.326
TriviaQA	0.924	0.056	0.105	0.227
CommonsenseQA	0.923	0.051	0.050	-0.037
SciQ	0.926	0.102	0.014	—
Average	**0.923**	0.024	-0.009	0.172

using DPO to align verbalized confidence with internal confidence. However, these approaches focus on explicit numerical confidence rather than natural hedging. Regarding human-AI interaction, Rathi et al. (2025) showed humans overrely on overconfident LLM generations across languages, and Zhou et al. (2024) introduced frameworks for measuring human reliance on LLMs.

3 Method

Given a question q, a model generates response r with associated token-level entropy values. We define hedging intensity $h(r) \in [0, 1]$ as uncertainty language measure and normalized entropy $e(r) \in [0, 1]$ as internal uncertainty. We train a neural hedging classifier combining learned features with rule-based pattern matching, blending neural predictions (55%) with rule-based scores (45%). For uncertainty, we compute entropy $H_t = -\sum_v p(v) \log p(v)$ per token, with normalized entropy $e(r) = \text{clip}(\log(1 + 1.5\bar{H}) / \log(1 + H_{\max}), 0, 1)$. We use REINFORCE with reward $R = \alpha \cdot R_{\text{correct}} + \beta \cdot R_{\text{align}} + \gamma \cdot R_{\text{corr}} + \delta \cdot R_{\text{div}}$, where $R_{\text{align}} = 1 - |h(r) - e(r)|$ rewards absolute alignment, R_{corr} rewards positive Spearman correlation, and R_{div} encourages hedging diversity. A KL penalty against a frozen reference model prevents policy collapse.

4 Experiments

We use GPT-2 (124M parameters) with only the last two transformer layers unfrozen, AdamW with learning rate 5×10^{-6}, temperature annealing from 0.9 to 0.5 over 8 epochs, and KL coefficient 0.01. We evaluate on Synthetic QA, TriviaQA (Joshi et al., 2017), CommonsenseQA (Talmor et al., 2019), and SciQ.

Table 1 presents our main results. UAH achieves uniformly high alignment scores (0.919–0.926) but correlations are consistently weak: Pearson r ranges from -0.11 to 0.10, and Spearman ρ from -0.20 to 0.10. The negative correlation on Synthetic QA is particularly concerning given this dataset was used for training. This disconnect between high alignment and weak correlation is our central negative finding.

Figure 1 reveals critical training behavior. The middle panel demonstrates that hedging-entropy correlation exhibits high volatility throughout training, initially dipping negative before recovering—this instability suggests the optimization landscape for correlation is fundamentally challenging. While alignment reaches 0.945 by epoch 8, the correlation only reaches approximately 0.55 during training but degrades substantially during evaluation (Table 1), revealing poor generalization of the learned hedging-entropy relationship.

Figure 2 provides direct evidence of calibration limitations. The clustering pattern reveals the model's failure mode: rather than tracking instance-level uncertainty, it outputs mid-range hedging values that minimize average absolute error. At high entropy (where hedging should be highest), points consistently fall below the diagonal, indicating the model fails to express appropriate uncertainty precisely when it matters most.

Figure 3 reveals the core limitation. The nearly flat calibration curve demonstrates the model learns a narrow hedging range regardless of entropy variation. By outputting values near the center of both distributions, the model minimizes absolute error without genuinely tracking uncertainty—explaining the disconnect between high alignment scores and weak correlations. Ablation studies

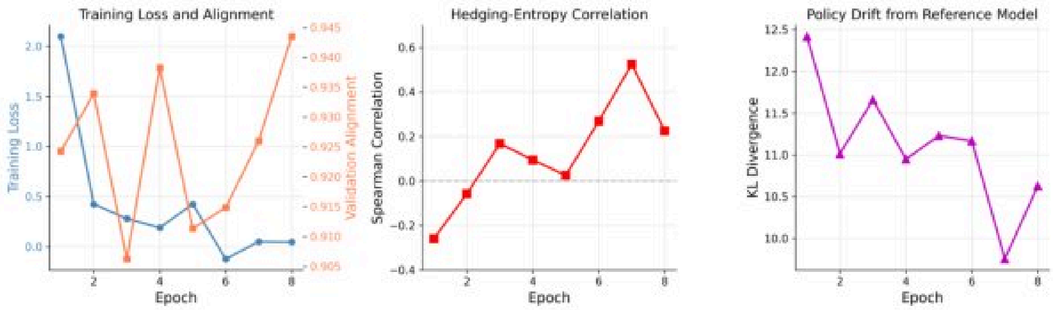

Figure 1: UAH training dynamics. Left: Training loss decreases while validation alignment improves to 0.945. Middle: Hedging-entropy correlation exhibits high volatility, dipping negative before recovering to ∼0.55—yet this degrades substantially during evaluation (Table 1). Right: KL divergence remains bounded (10–12.5), indicating controlled policy drift.

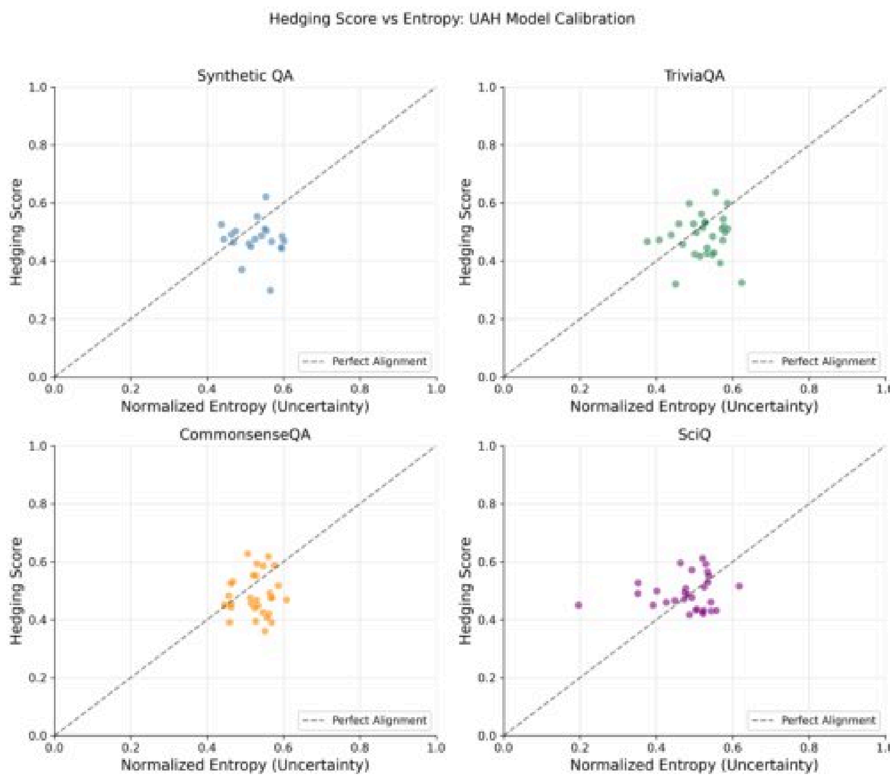

Figure 2: Hedging versus entropy across datasets. Points cluster between entropy 0.3–0.8 and hedging 0.3–0.7. The majority fall below the perfect calibration diagonal at higher entropy values, revealing systematic under-hedging when the model is most uncertain.

(Appendix A) show removing the diversity bonus severely reduces peak correlation (0.52 → 0.28), confirming hedging collapse as a major failure mode.

5 Conclusion

Our experiments reveal a fundamental challenge in teaching language models to communicate uncertainty naturally. While UAH achieves high alignment scores, this metric proves insufficient for genuine calibration. The model learns to output mid-range hedging values that score well on average but fail to track instance-level uncertainty. Contributing factors include discretized classifier

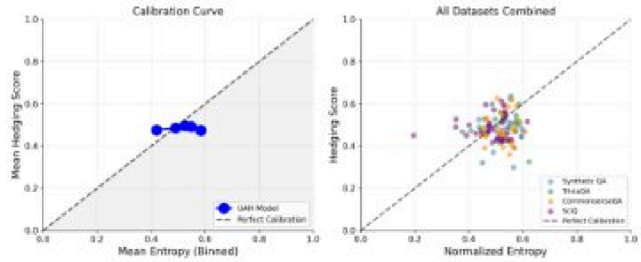

Figure 3: Calibration analysis. Left: The calibration curve is nearly flat (hedging 0.47–0.55) despite entropy varying 0.40–0.60, rather than following the diagonal. Right: Combined scatter showing the overall under-hedging pattern.

outputs limiting training signal dynamic range, token-level entropy potentially missing semantic uncertainty, and the alignment objective rewarding regression to the mean. Future work should explore semantic entropy measures, richer hedging representations, and evaluation frameworks capturing human-relevant calibration.

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Supplementary Material

A Ablation Analysis

Table 2 reveals that removing the correlation reward term actually increases peak Spearman correlation (0.52 → 0.67), suggesting potential interference between reward components. The diversity bonus proves critical—its removal causes hedging collapse with peak correlation dropping to 0.28.

Table 2: Ablation study results showing component contributions.

Configuration	Alignment	Peak Spearman	Min KL
Full UAH	0.923	0.52	9.8
No KL Penalty	0.921	0.48	10.7
No Diversity Bonus	0.918	0.28	10.1
No Correlation Reward	0.925	0.67	10.0
Rule-Based Only	0.790	0.43	9.9

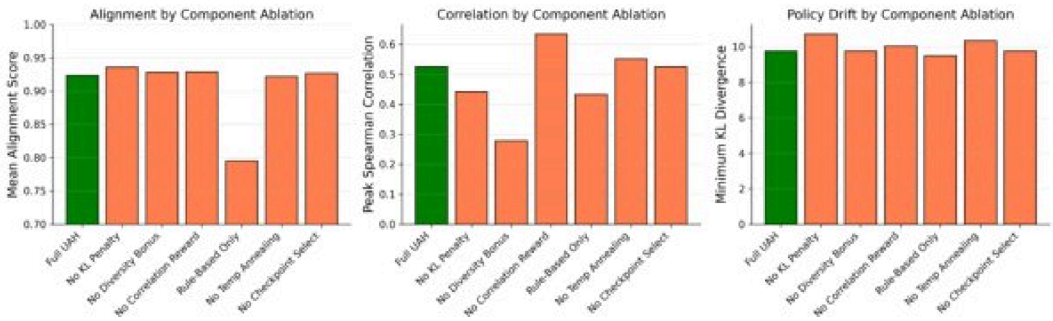

Figure 4: Ablation results across configurations. Left: Alignment remains high (>0.91) except for rule-based-only. Middle: Peak correlation varies substantially—removing correlation reward achieves highest correlation (0.67), while removing diversity causes collapse (0.28). Right: KL divergence remains stable across ablations.

B Baseline Experiments

Figure 5 shows a critical dissociation: top_p=0.85 achieves the lowest training loss but produces the worst validation alignment, collapsing to negative values. This demonstrates that standard fine-tuning metrics do not predict uncertainty calibration quality.

C Cross-Dataset Generalization

D Training Hyperparameters

Table 3: Training hyperparameters for UAH experiments.

Parameter	Value
Base Model	GPT-2 (124M parameters)
Trainable Layers	transformer.h.10, transformer.h.11, lm_head
Learning Rate	5×10^{-6}
Optimizer	AdamW
Training Epochs	8
Batch Size	4
Temperature (initial/final)	0.9 / 0.5
Top-p	0.9
KL Coefficient	0.01
Reward Weights	
Correctness (α)	0.35
Alignment (β)	0.35
Correlation (γ)	0.15
Diversity (δ)	0.15

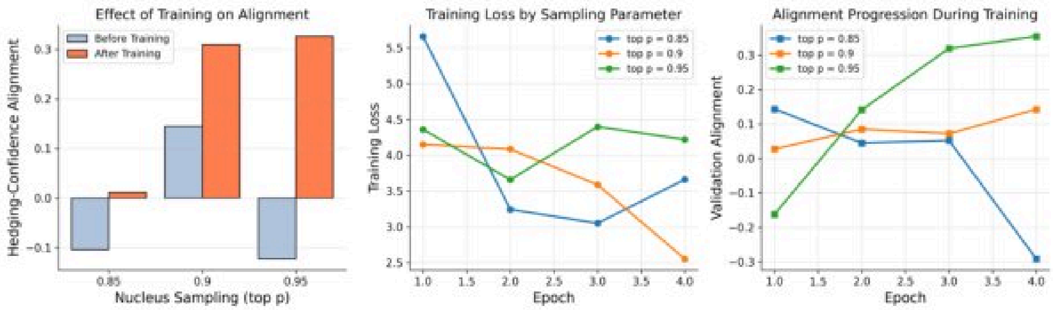

Figure 5: Baseline tuning across nucleus sampling parameters. Top_p=0.85 achieves lowest training loss but produces worst validation alignment (collapsing to negative values), demonstrating that standard fine-tuning metrics do not predict uncertainty calibration quality.

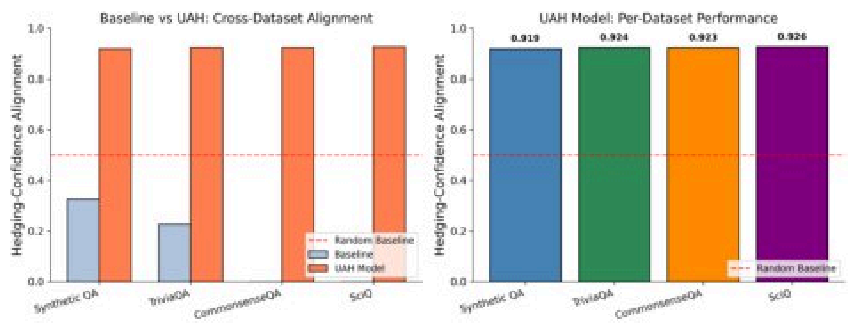

Figure 6: Cross-dataset comparison. UAH achieves consistent alignment (0.919–0.926) across all datasets, while baseline performance varies from -0.037 to 0.326. However, this consistency indicates regression to the mean rather than genuine calibration.

E Hedging Classifier and Dataset Details

The neural hedging classifier uses a two-layer MLP with 64 and 16 hidden units. Hedging patterns are organized into three tiers: high-hedge (weight 0.9: "I'm not sure," "maybe"), medium-hedge (weight 0.5: "I think," "probably"), and certainty markers (weight -0.3: "definitely," "certainly"). The Synthetic QA dataset contains 500 questions across 5 difficulty levels. TriviaQA (Joshi et al., 2017), CommonsenseQA (Talmor et al., 2019), and SciQ each contribute 200 sampled questions for evaluation.